

Perplexity Improves While MMLU Lags: A Depth Study of Prune-then-Heal Language Models

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Abstract

Depth-pruned language models are commonly continued-pretrained to restore aggregate language-model quality, but lower perplexity does not by itself identify which behaviors have returned. We present a trajectory-level study of post-pruning recovery on OLMo-2 base models. Prefix variants retain pretrained lower blocks, append a fresh tail, and continue pretraining; we evaluate held-out likelihood, two MMLU scoring interfaces, three closed-book QA tasks, and a broader likelihood suite. In the 16-layer keep14 arm, perplexity improves to 10.561 after 200k steps but remains $1.428\times$ the intact model’s 7.398, while answer-letter MMLU reaches only .319 versus .605. An intact 32-layer control stays near the base through its observed 25k plateau checkpoint, making short-horizon corpus shift an insufficient explanation. Complete-option scoring raises keep14 to .383, consistent with an answer-interface component; however, a fully random 16-layer model reaches nearly the same content score while remaining at chance on answer letters, exposing a large fluency floor. The deficit also persists on PopQA, TriviaQA, and NQ-open. A different construction, non-contiguous ShortGPT-16, reaches .474 MMLU at the same 200k target budget, although coupled structural differences prevent attribution to a single retained block or selection rule. Our conclusion is methodological: after a structural intervention, likelihood, target capability, scoring interface, exact construction, and recovery compute should be reported separately; perplexity measures distributional fit, not a sufficient certificate of capability recovery.

1 Introduction

Removing transformer blocks and then continuing pretraining (CPT, often called “healing”) is an attractive route to smaller language models (Gro-
mov et al., 2025; Kim et al., 2024; Men et al., 2024; Yang et al., 2024). A pruned model can improve

rapidly on held-out next-token loss, and endpoint perplexity is therefore a natural measure of whether retraining is working. It is not, however, a complete description of what the structural intervention removed or what CPT restored.

Perplexity averages local prediction loss over a corpus. Factual recall, multiple-choice answer selection, generation, and task-specific interfaces probe different observables. Compression studies have documented that aggregate language-model metrics can miss changes in parametric knowledge, downstream behavior, or safety (Namburi et al., 2023; Xu et al., 2024; Jaiswal et al., 2024). Yet most depth-pruning work reports a small number of endpoints. Endpoint comparisons cannot distinguish slow recovery from a stable deficit, corpus drift from structural damage, or competence loss from failure to express an answer through a particular scoring interface.

Research question. We ask: *what does continued pretraining restore after depth pruning, at what observed rate, and which controls are required to interpret the result?* We treat recovery as a multi-axis object—held-out likelihood, target capability, scoring interface, structural policy, and training budget—rather than a scalar certified by perplexity.

We instantiate the study on OLMo-2-1124-7B (OLMo Team et al., 2025). Prefix arms retain k pretrained decoder blocks, append two randomly initialized blocks, and continue pretraining on a fixed Dolmino stream. The principal keep14 arm is tracked to 200k steps. Controls include the intact base, an unpruned model continued on the same corpus array and nominal schedule to a pre-registered 25k plateau checkpoint, a frozen-prefix arm, a fully random same-shape model, and a non-contiguous ShortGPT-16 endpoint. Evaluation combines held-out likelihood, a broad zero-shot likelihood suite, answer-letter and complete-option MMLU, and no-retrieval PopQA, TriviaQA, and

Prune-regrow-heal exposes a multi-axis recovery gap

OLMo-2-1124-7B · endpoint values at 200k healing steps

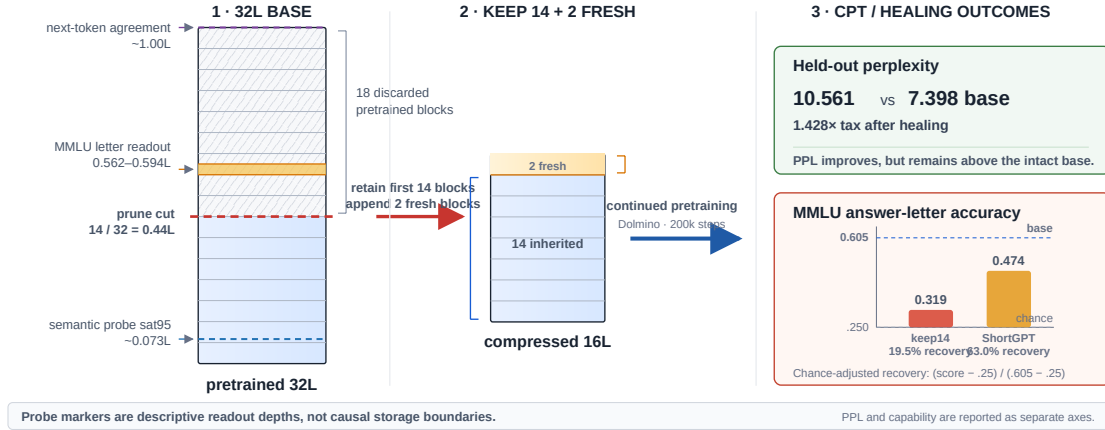


Figure 1: **Post-pruning recovery is multi-axis.** The principal recipe keeps 14 of 32 pretrained OLMo-2 blocks, appends two fresh blocks, and continues pretraining. At 200k steps, held-out PPL improves sharply but remains $1.428\times$ the base, while answer-letter MMLU recovers only 19.5% of the base’s above-chance margin. A different 16-layer construction, ShortGPT, reaches 63.0%, showing that nominal depth does not determine one endpoint. Layer-wise readout markers are descriptive correlates, not causal storage boundaries.

NQ-open.

The central observation is bounded but robust across interfaces. keep14 reaches PPL 10.561, yet remains far below the intact model on answer-letter MMLU and all three closed-book tasks. Complete-option scoring narrows the MMLU gap, consistent with an answer-symbol or readout component, but the random-init content-score floor rules out interpreting that protocol as recovered subject knowledge. ShortGPT is substantially stronger on both PPL and MMLU, so the keep14 deficit is not a universal consequence of using 16 blocks. Coupled structural differences prevent a single-factor explanation.

Contributions.

- **Trajectory-level measurement.** We track likelihood and knowledge-sensitive behavior during post-depth-pruning CPT rather than evaluating only a final compressed endpoint.
- **Controls that separate interpretations.** Continued-full-model, same-shape initialization/adaptation, scoring-interface, policy, and closed-book controls distinguish corpus drift, fluency floors, answer format, and structural choice.
- **A reporting protocol.** We show why post-pruning likelihood, target capability, interface, exact structure, and recovery compute should be reported as separate axes. Layer-wise readouts

provide a compatible but explicitly non-causal correlate.

2 Related Work

We do not propose a new pruning criterion. Our object of study is the recovery trajectory after a structural intervention and the controls needed to interpret it. Table 1 gives a compact positioning.

Selecting and removing depth. ShortGPT and SLEB rank and remove redundant blocks (Men et al., 2024; Song et al., 2024); LaCo merges adjacent layers (Yang et al., 2024); Block-Pruner searches at attention/MLP-block granularity (Zhong et al., 2025). Other studies show that preferred removals depend on task and calibration choices (Siddiqui et al., 2024; Lu et al., 2024; Kim et al., 2026). These works optimize *which* structure to retain. We instead measure what different retained structures recover under CPT.

Retraining pruned models. Gromov et al. (2025) remove contiguous deeper layers and show that post-healing autoregressive loss and downstream tasks can behave differently. Our narrower increment is to treat the recovery path itself as the analysis object and add same-corpus, same-shape, scoring-interface, construction, and closed-book controls. Shortened LLaMA directly compares CPT and LoRA after depth pruning (Kim et al., 2024). Sheared LLaMA, Minitron, compact-model distillation, and DRPruning combine pruning with

Study family	Intervention	Adaptation	Reported recovery	Trajectory?	Key control(s)
ShortGPT / SLEB	ranked block removal	none	PPL + task endpoints	no	importance/policy variants
Gromov et al.	contiguous deep-block removal	continued pretraining	post-heal PPL + tasks	limited	depth and healing budget
Sheared / compact / DRPruning	structured pruning	CPT or distillation	compact-model endpoints	limited	data/training recipe
This study	prefix+fresh tail; ShortGPT endpoint	controlled healing	PPL + task + interface	yes	full32, init, freeze, policy

Table 1: **Positioning of the study.** This study does not introduce a new importance score. Its distinguishing object is the recovery trajectory, together with controls that separate structural intervention, continued-training drift, initialization/adaptation, pruning policy, and scoring interface. “Limited” means that intermediate training points are not the primary analysis object.

data allocation or teacher supervision (Xia et al., 2024; Sreenivas et al., 2024; Muralidharan et al., 2024; Deng et al., 2025). Our prefix-plus-fresh-tail recipe is not proposed as a superior compression algorithm; it is a controlled setting in which to track recovery and contrast structural policies.

Evaluating compression beyond perplexity.

The Cost of Compression studies how pruning and quantization affect parametric knowledge (Namburi et al., 2023). Jaiswal et al. (2024) show that low perplexity can coexist with large deficits on knowledge-intensive compressed-LM benchmarks, and Xu et al. (2024) document divergent safety effects. These studies motivate multi-dimensional evaluation. Our additional focus is temporal and causal-inference discipline: within-arm checkpoints, an intact same-corpus control, same-shape operating points, scoring-interface decomposition, and a construction comparison.

Readout across depth. Logit-lens and tuned-lens analyses decode intermediate predictions (nostalgebraist, 2020; Belrose et al., 2023); feed-forward memory, knowledge-neuron, and causal-tracing work connect particular computations to factual behavior (Geva et al., 2021; Dai et al., 2022; Meng et al., 2022). DoLa and LayerSkip exploit differences in layer-wise predictions for decoding or early exit (Chuang et al., 2024; Elhoushi et al., 2024). We use readout depth only as a descriptive correlate. Neither a probe transition nor pruning damage identifies where knowledge is stored.

3 Prune–Regrow–Heal Study

3.1 Intervention and model family

We study OLMo-2-1124-7B (OLMo Team et al., 2025), a 32-layer base language model without instruction tuning. For prefix arm $\text{keep} = k$, we retain the first k pretrained decoder blocks and append two randomly initialized blocks. We denote the cut location by $d_{\text{cut}} = k/32$ and resulting depth

by $d_{\text{model}} = (k + 2)/32$. Embeddings, final normalization, and the output head are copied exactly. The 1B OLMo-2 model provides a qualitative same-family replication.

The principal arm, keep14, is therefore a 16-layer model containing 14 inherited prefix blocks and two fresh tail blocks. We additionally evaluate an influence-selected ShortGPT-16 endpoint that inherits 16 non-contiguous blocks, including the original final block. It is a policy comparison, not a selection-only ablation, because inherited count and fresh-tail construction also differ.

3.2 Healing

All inherited prefix arms continue pre-training on the DCLM portion of dolmino-mix-1124 (Soldaini et al., 2024), retokenized into 2,048-token windows. A disjoint 4096×2048 shard provides held-out likelihood. The maximum budget is 200k optimizer steps with effective batch size 128. In inherited runs, all trainable parameters use peak learning rate 2×10^{-5} ; AdamW, cosine decay, mixed precision, and reconstruction checks are detailed in Appendix B.

keep14 and ShortGPT complete 200k steps. Shallower prefix runs are evaluated at their retained plateau checkpoints (keep8 121k, keep10 83.5k, and keep12 124k) after their knowledge-sensitive metrics had stabilized; we report their literal checkpoint steps and do not treat them as compute-matched 200k measurements.

3.3 Control branches

Four controls delimit interpretation. The intact 32-layer model is continued on the same corpus array and nominal schedule, measuring short-horizon training drift without pruning. Frozen-front keep14 trains the fresh tail and non-block modules without adapting inherited blocks. A fully random 16-layer model matches shape, data, batch, and maximum steps but uses a higher learning rate, so it is an

operating point rather than an initialization-only ablation. Finally, ShortGPT provides a different construction at the same resulting depth. The study does not isolate its coupled structural differences, and current claims are limited accordingly.

3.4 Evaluation interfaces

Held-out perplexity is raw next-token loss with no chat template or inserted BOS. The principal downstream suite uses zero-shot likelihood scoring. Standard MMLU (Hendrycks et al., 2021) shows all four options and scores the single answer letter. A paired content protocol instead scores each complete option text, reporting both summed and per-token mean log-likelihood. Because a random-init model obtains a substantial normalized-content score, we interpret this protocol as an interface diagnostic rather than a direct knowledge measure.

We additionally evaluate PopQA (Mallen et al., 2023), TriviaQA (Joshi et al., 2017), and NQ-open (Kwiatkowski et al., 2019) with a shared zero-shot, no-retrieval Question: /Answer: generation protocol. The broader likelihood suite covers HellaSwag, ARC, PIQA, WinoGrande, LAMBADA, BoolQ, OpenBookQA, CommonsenseQA, and SocialQA (Zellers et al., 2019; Clark et al., 2018; Bisk et al., 2020; Sakaguchi et al., 2021; Parnano et al., 2016; Clark et al., 2019; Mihaylov et al., 2018; Talmor et al., 2019; Sap et al., 2019). Metrics, sample counts, prompts, checkpoint provenance, and strict architecture reconstruction appear in Appendix B.

4 Recovery as a Multi-Axis Measurement

Perplexity aggregates token-level negative log-likelihood over a corpus. A knowledge-sensitive task instead asks whether a particular answer can be selected or generated under a task interface. CPT can therefore improve the former without changing the latter at the same observed rate. This does not make perplexity uninformative; it means that distributional fit is not a certificate for every behavior after a structural intervention.

This distinction motivates three study choices. First, we retain within-arm checkpoints because a single endpoint cannot distinguish delayed from absent change over the observed budget. Second, answer-letter and complete-option scoring are reported separately because they mix competence and readout differently. Third, we compare structural constructions because the same nominal depth

need not imply the same inherited computation. These are measurement requirements, not a causal theory of where capability resides.

Layer-wise probes provide supplementary context: on intact OLMo-2-7B, frozen semantic classifiers become readable early, MMLU answer-letter readout rises near layers 18–19, and exact next-token agreement saturates at the top. Because these quantities use different probes and thresholds, we treat their ordering as descriptive only; full results are in Appendix C.

5 Experiments

5.1 Experimental setup

All main results use OLMo-2-7B under the base-model protocol in §3. The central comparisons are the intact base, full32 continued-pretraining control, keep14+fresh2, ShortGPT-16, frozen-front, and fully random-init operating points. Detailed per-task results and all available checkpoints are in Appendix A.

5.2 Main results: likelihood improves faster than knowledge-sensitive evaluation

Table 2 gives the central result. The half-depth keep14 arm reaches PPL 10.561 after 200k steps—a large improvement, but still $1.428\times$ the base—and remains far below the base on both MMLU interfaces and all three closed-book tasks. Short-horizon corpus shift is insufficient to explain the gap: the intact full32 control remains near the base at its observed step-25k plateau checkpoint in held-out likelihood, MMLU, and closed-book evaluation.

The MMLU interface matters but does not erase the result. Scoring complete option text raises keep14’s above-chance recovery relative to answer-letter scoring, which is consistent with a content-to-symbol or readout contribution. Yet fully random initialization attains nearly the same normalized-content score as frozen-front despite chance-level answer letters. The content protocol therefore contains a large fluency floor; neither interface alone is identified with stored knowledge.

The separation is not specific to multiple choice. At 200k, keep14 reaches only .142 PopQA answer containment, .294 TriviaQA exact match, and .060 NQ-open exact match, compared with .257, .636, and .205 for the intact base. These factual recall tasks use a shared no-retrieval generation interface, while full32 continued pretraining preserves much

Operating point	PPL↓	MMLU letter	MMLU content	PopQA	TriviaQA	NQ-open
Full base (32L)	7.398	.605	.471	.257	.636	.205
Full32 continued@25k [†]	7.670	.588	.466	.228	.572	.158
keep14 + fresh2, 200k	10.561	.319	.383	.142	.294	.060
ShortGPT-16, 200k	9.780	.474	.401	—	—	—
Frozen inherited front, 200k	12.797	.262	.360	.128	.248	.050
Fully random init, 200k	11.498	.247	.360	.111	.209	.063

Table 2: **Likelihood and knowledge-sensitive evaluation recover differently.** MMLU letter scores the answer symbol; MMLU content scores the complete option text using mean log-likelihood per continuation token, the pre-specified content headline. PopQA reports normalized-answer containment; TriviaQA and NQ-open report exact match. Full32 continued pretraining stays near the base, whereas every tested half-depth operating point remains substantially below it. The elevated content score of fully random initialization exposes a fluency floor, so content accuracy is not interpreted as recovered knowledge. [‡]The separately rerun interface table reports letter accuracy to four decimals (.3184 for keep14 and .6054 for the base); the rounded canonical benchmark cells here use .319 and .605. [†]All full32 cells use the same intact 32-layer step-25k plateau checkpoint; “25k” is the actual checkpoint label, not 200k.

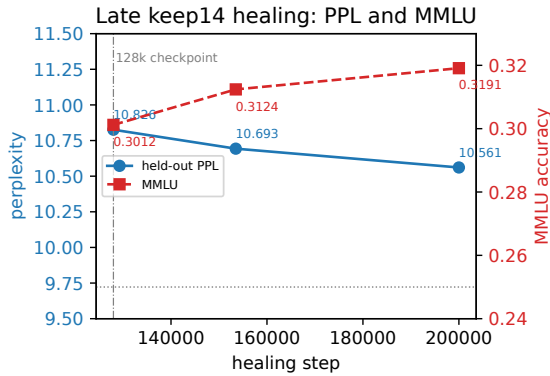


Figure 2: **Late keep14 healing continues on both axes, but the remaining capability gap is large.** From 128k to 200k, PPL improves from 10.826 to 10.561 and MMLU from .3012 to .3191; the intact base remains at .6053 MMLU. The figure supports slow late recovery, not a claim that MMLU is numerically constant.

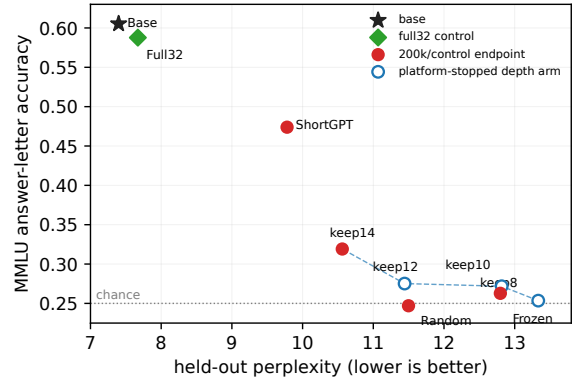


Figure 3: **Held-out PPL does not uniquely determine answer-letter MMLU across interventions.** keep14, ShortGPT, frozen-front, and random-init are 200k points; shallower prefix rows use their retained plateau checkpoints. Filled red points are controls or ShortGPT; the dashed line connects the prefix ladder as a visual guide.

more of the base performance. Together with the two MMLU protocols, they rule out an answer-letter-only explanation.

Figure 3 makes the measurement issue visible. PPL and MMLU broadly co-vary across the strongest endpoints, but no scalar PPL threshold certifies capability: the random-init arm has lower PPL than frozen-front yet lower MMLU, while several structurally distinct endpoints cluster near chance. The plot therefore motivates reporting both axes; it is not evidence that they are globally uncorrelated.

5.3 Controlled comparisons

Same-shape initialization and adaptation. At 200k, keep14, frozen-front, and fully random init share the 16-layer shape. Their metric rankings disagree: frozen-front has worse PPL than random init but higher answer-letter MMLU, while train-all keep14 is strongest on both. Paired item-level tests confirm the keep14 advantage over each control, but learning rate and trainable-module differences prevent an initialization-only or adaptation-only causal claim. Appendix Table 15 reports exact McNemar tests and paired-bootstrap intervals.

Full-model corpus control. Through the observed 25k plateau checkpoint, continuing the intact 32-layer model leaves PPL and both MMLU interfaces close to the original base and preserves most PopQA, TriviaQA, and NQ-open performance. Thus short-horizon corpus shift is insufficient to explain the large half-depth gap. This control is not step- or token-exposure-matched to keep14 at 200k, so we do not treat it as an exact long-budget counterfactual.

Scoring-interface control. Across all nine evaluated arms, complete-option MMLU is less depth sensitive than answer-letter MMLU. ShortGPT remains stronger than the prefix arms under both interfaces, showing that the observed endpoint differs across constructions. The full table and protocol definition are in Appendix Table 16.

6 Analysis

6.1 Recovery varies across knowledge domains

Figure 4 separates absolute accuracy from recovery relative to each group’s base margin. keep14 is strongest on the heterogeneous Other group (.377; 29.1% chance-adjusted recovery) and weakest on STEM (.294; 15.6%) and Humanities (.299; 16.2%). Social Science has higher raw accuracy (.335) but only 18.6% recovery because its base is much stronger. ShortGPT improves every group, most notably Other (.574) and Social Science (.554), yet still remains below full32.

Subject-level examples show that the pattern is not a single-domain artifact. Among high-sample subjects, keep14 retains relatively more on miscellaneous ($n=783$: .457 versus .803 base) and marketing ($n=234$: .440 versus .825), but little on professional law ($n=1,534$: .272 versus .460) and high-school statistics ($n=216$: .213 versus .514). These examples are descriptive: group membership mixes base difficulty, answer format, corpus exposure, factual recall, and reasoning. They do not establish that a particular knowledge type is stored in the removed blocks.

6.2 Prefix depth interacts with observed healing

keep8 and keep14 provide complementary within-arm trajectories. keep8’s PPL falls smoothly, but a paired 45k–121k comparison finds no significant MMLU change (+0.24 points, 95% CI $[-0.51, 0.98]$). keep14 instead gains 1.68 MMLU

points from 128k to 200k (95% CI $[1.08, 2.29]$) while PPL improves by 0.265 (Figure 2). The gain is statistically detectable but leaves a large gap to the base. This is evidence that observed recovery rate depends on inherited depth within the prefix recipe; it is not a claim about eventual convergence under unlimited compute.

6.3 What makes ShortGPT different?

ShortGPT and keep14 end at 16 blocks and complete the same 200k target budget, but differ along four coupled dimensions: ShortGPT inherits 16 rather than 14 pretrained blocks, selects them non-contiguously, retains original block 31, and uses no fresh tail. keep14 retains a contiguous prefix and appends two randomly initialized blocks. Embeddings, final normalization, and output heads are copied from the same base and therefore do not explain the difference.

The resulting endpoint gap is large: ShortGPT reaches .474 MMLU versus .319 for keep14 while also obtaining lower PPL. Thus nominal depth alone does not determine the observed endpoint across these constructions. It does *not* show that block 31 is uniquely responsible, that non-contiguous selection is optimal, or that fresh tails are harmful. Additional matched structures would be needed to test these hypotheses separately.

6.4 Generality beyond the principal OLMo-2 setting

A more compressed OLMo-2-1B arm exhibits the same qualitative pattern of falling PPL with near-chance MMLU, providing same-family context rather than an independent replication. An existing Qwen3-8B endpoint supplies a limited cross-family check: a 14/36-layer prefix+fresh-tail model remains at .2495 MMLU after 200k steps, compared with .7297 for its base (Appendix Table 9). Qwen retains a smaller depth fraction than the OLMo keep14 recipe (39% versus 44%), which is one plausible contributor to its larger deficit, but the corpus, architecture, and missing content/closed-book controls prevent attribution. We therefore treat Qwen as directional evidence, not a matched replication.

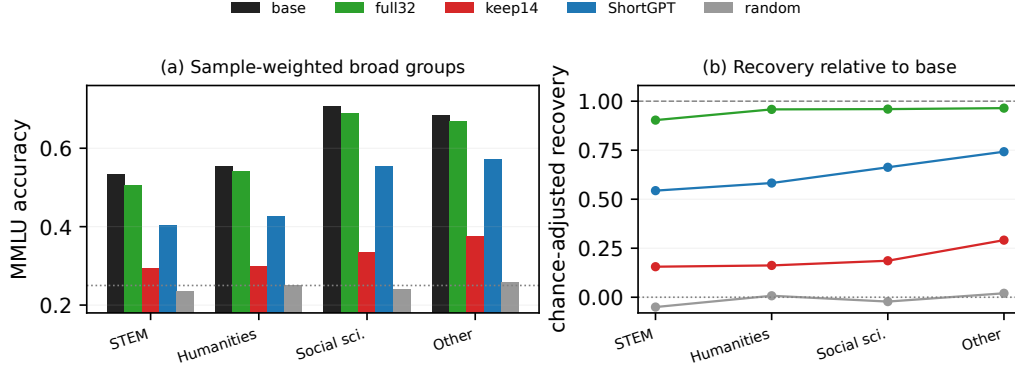


Figure 4: **MMLU recovery is heterogeneous across broad domains.** Accuracy is sample weighted over the standard four MMLU groups. Recovery is chance adjusted, $(a - 0.25)/(a_{\text{base}} - 0.25)$. Full32 remains close to the base in every group; ShortGPT consistently exceeds keep14, but neither half-depth arm closes the intact-model gap.

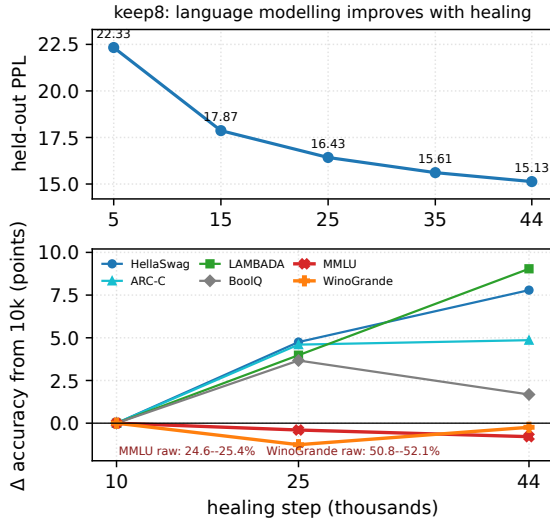


Figure 5: **The shallow keep8 arm improves in likelihood and ordinary tasks through the early 44k trajectory.** The separate paired 45k–121k comparison in the text finds no detectable MMLU change.

Arm	Step	Inh./fresh	PPL↓	core6	MMLU
full base (32L)	—	32/0	7.398	.7037	.6053
keep8+fresh2 (10L)	121k	8/2	13.333	.5238	.2535
keep10+fresh2 (12L)	83.5k	10/2	12.816	.5303	.2718
keep12+fresh2 (14L)	124k	12/2	11.443	.5669	.2752
keep14+fresh2 (16L)	200k	14/2	10.561	.5938	.3191
ShortGPT-16	200k	16/0	9.780	.6215	.4739

Table 3: **Available depth and policy endpoints.** core6 averages six commonsense/reasoning tasks; Appendix Table 10 gives all cells. ShortGPT retains layers [0–12, 16, 17, 31] and matches keep14 in total depth and completed 200k training, but differs in inherited count and fresh-tail construction. The prefix ladder is not compute matched: shallower runs use their retained plateau checkpoints, whose PPL was still decreasing.

7 Discussion

What does healing restore? The combined evidence argues against treating recovery as a single scalar. PPL measures how well the healed model fits held-out next-token events; it improves substantially in every inherited-prefix arm. That improvement is useful—the models recover fluent continuation and several commonsense tasks—but it does not uniquely determine answer selection or closed-book recall. The endpoint scatter makes the distinction concrete: models with similar PPL can occupy meaningfully different MMLU regimes, and random initialization can outrank a frozen inherited front in PPL while underperforming it on answer letters. We therefore interpret healing as partial recovery of distributional modeling plus a task-dependent remainder, not as a binary return of the original model.

Interface sensitivity is substantial but insufficient. Complete-option MMLU recovers more smoothly than answer-letter MMLU, showing substantial protocol sensitivity that is consistent with an answer-symbol component. The protocols also change the prompt, candidate continuation, tokenization, and normalization, so they do not isolate mapping loss. The random-init fluency floor and the independent PopQA, TriviaQA, and NQ-open gaps rule out a formatting-only account. This distinction matters for evaluation: a single multiple-choice protocol can overstate damage when its interface is brittle, whereas a content-likelihood protocol can overstate recovery when generic fluency makes plausible options easy to score. Agreement across interfaces is more informative than designating either as ground truth.

Different constructions yield different endpoints. The step-matched keep14–ShortGPT comparison shows that half depth is not one operating point. Preserving selected upper blocks yields a much stronger likelihood–capability endpoint than retaining only a prefix and regrowing a tail. The result does not identify whether the advantage comes from the original final block, two additional inherited blocks, non-contiguous selection, or avoiding fresh layers. It shows that conclusions drawn from one prefix construction should not be generalized to every 16-layer construction; isolating the responsible factor requires additional matched structures.

Which behaviors remain difficult? The residual deficit spans factual recall (PopQA, TriviaQA,

and NQ-open), broad academic knowledge (all four MMLU groups), and some answer-interface behavior. In contrast, several continuation-style commonsense tasks recover more readily. This taxonomy is behavioral rather than anatomical. Knowledge-neuron and causal tracing studies motivate asking whether particular feed-forward computations support factual recall (Dai et al., 2022; Meng et al., 2022), but neither our subject groups nor ShortGPT’s retention of block 31 identifies the missing substrate. Structure-isolation and causal interventions are required before making that claim.

A reporting protocol for structural compression. For prune-then-heal studies, we recommend reporting likelihood, target capability, recovery compute, and exact structural policy together. An intact model continued on the same corpus and nominal schedule measures short-horizon drift, while same-shape initialization/adaptation controls expose fluency floors and inherited-computation effects.

Before claiming recovery from PPL alone: (1) evaluate an independent target capability; (2) test a second scoring or generation interface; (3) include a continued-full-model corpus control; and (4) report the exact structure and healing budget.

This checklist does not replace efficiency or endpoint-quality comparisons. It prevents a low PPL alone from being interpreted as evidence that the behaviors of interest have returned.

8 Conclusion

Post-pruning recovery is not one-dimensional. In the observed OLMo-2 prefix recipe, held-out likelihood improves substantially while answer-letter MMLU and closed-book QA remain far below the intact model. An intact same-corpus control, complete-option scoring, same-shape operating points, and a ShortGPT construction comparison show that corpus drift, answer interface, initialization/adaptation, and exact structure each affect the interpretation. Perplexity remains an essential measure of distributional fit, but it is not a sufficient certificate that a target behavior has returned after structural intervention. Compression studies should report likelihood, target capability, scoring interface, exact structure, and recovery compute as separate axes.

Limitations

This is a measurement study of one principal OLMo-2-7B pruning recipe, not a new state-of-

the-art compression method. keep14 and the same-shape controls are single training runs; item-level intervals quantify evaluation uncertainty, not training-seed variance. Shallower prefix arms stop at retained plateau checkpoints rather than a common 200k step, and the intact full32 control stops at 25k. Consequently, the depth ladder and full32 comparison are not step-, token-exposure-, or FLOP-matched long-budget counterfactuals.

The fully random model uses a higher learning rate, while frozen-front changes the trainable parameter set. They are informative operating points rather than clean initialization or adaptation ablations. ShortGPT differs from keep14 in inherited block count, non-contiguous selection, final-block retention, and the absence of a fresh tail; its advantage cannot be attributed to any one of those factors. It shows only that the two coupled constructions yield different endpoints.

Answer-letter and complete-option MMLU use different prompts and candidate scoring. Their gap is consistent with an interface component, but the random-init fluency floor prevents content accuracy from being interpreted as subject knowledge. Conversely, closed-book exact match depends on generation and normalization. We therefore rely on agreement across interfaces and datasets rather than designating one metric as ground truth.

Layer-wise logit-lens and linear-probe thresholds are readout dependent. Their ordering is descriptive and neither localizes factual storage nor proves that removed layers contained the missing capability. The study covers English base models, one healing mixture, and primarily 7B scale; instruction tuning, other architectures, pruning policies, corpora, and longer budgets may produce different recovery trajectories.

Held-out PPL is measured on a disjoint shard from the same Dolmino/DCLM source as healing. We do not report out-of-domain PPL or a benchmark-contamination audit, so the likelihood axis should be interpreted as in-domain distributional fit and the knowledge-sensitive results as observed benchmark behavior, not a clean estimate of newly recovered world knowledge.

Ethical Considerations

This work studies how model compression via layer pruning and continued pretraining affects evaluated capability; it does not add a safety mechanism and changes none of the underlying model’s

risks such as hallucination, biased generation, or unsafe completion. The observed PPL–MMLU dissociation carries a practical caution: a compressed model that looks healthy on aggregate language-modelling metrics may still perform poorly on a target benchmark, and deployments should evaluate the capabilities they intend to preserve rather than rely on perplexity alone.

We collect no new human-subject data and recruit no annotators. All experiments use previously released artifacts—the OLMo-2 base model, the Dolmino corpus, and standard public benchmarks—under their respective licenses. Training and large-scale evaluation consume energy. We reuse a common training recipe and report checkpoint step explicitly rather than presenting unequal-step results as compute matched. Any release should preserve the licenses and usage restrictions of the underlying model, corpus, and benchmarks.

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A Complete Results

The main paper emphasizes the observed-budget perplexity–MMLU dissociation. This appendix reports all consolidated checkpoints, including the qualitative 1B comparison, within-arm trajectories, and chance-adjusted summaries.

A.1 Held-out perplexity checkpoints

Table 4 distinguishes progressing checkpoints from true endpoints. The fully random-init arm is a random language model with the same 16-layer architecture: its decoder blocks, embedding, final norm, and output head are initialized without OLMo-2 weights. It uses a different learning rate, so it is a budgeted operating point rather than an initialization-only ablation. Table 5 reports both immediate damage and the healed non-contiguous endpoint. Its 16 retained layers are selected by cosine block influence on 128 Dolmino windows and include the original final layer.

A.2 Within-architecture training trajectories

The complete 11-task keep8 trajectory in Table 6 is the cleanest 7B test of uniform under-training. Across a $4.4\times$ increase in optimization steps, nine reported tasks improve by 1.7–9.0 accuracy points, while MMLU stays within 0.79 point of chance and WinoGrande varies by only 1.26 points near chance. This rejects a uniform-delay account over the observed interval, but not recovery after substantially more training.

The OLMo-2-0425-1B arm shows the same qualitative separation (Figure 6; Tables 7–8): PPL falls $17.619 \rightarrow 15.628$ and HellaSwag/ARC-Easy/PIQA improve, while MMLU is .2495/.2558/.2529/.2480. Because this is one more-compressed arm from the same model family with a weaker base, we treat it as qualitative context, not an independent replication.

A.3 Capability recovery and continued healing

Table 11 exposes why a single average benchmark score would be misleading. Fully random initialization nearly matches inherited keep14 on ARC-Easy, PIQA, and BoolQ, but inherited initialization retains more signal on WinoGrande, MMLU, and several other downstream tasks. The late-healing comparison in Table 13 shows no broad rollback through 200k: from 153.5k to 200k, PPL improves by 0.132 while MMLU gains only 0.67 points and

Scale	Arm	step	inherited	total layers	PPL↓	tax↓
7B	full base	—	32	32	7.398	1.000×
	keep8+fresh2	5k	8	10	22.331	3.019×
	keep8+fresh2	15k	8	10	17.868	2.416×
	keep8+fresh2	25k	8	10	16.426	2.220×
	keep8+fresh2	35k	8	10	15.612	2.110×
	keep8+fresh2	44k	8	10	15.131	2.045×
	keep8+fresh2	121k [†]	8	10	13.333	1.802×
	keep10+fresh2	10k	10	12	17.239	2.330×
	keep10+fresh2	83.5k [†]	10	12	12.816	1.732×
	keep12+fresh2	111.5k	12	14	11.566	1.563×
	keep12+fresh2	124k [†]	12	14	11.443	1.547×
	keep14+fresh2	128k	14	16	10.826	1.463×
	keep14+fresh2	153.5k	14	16	10.693	1.445×
	keep14+fresh2	200k	14	16	10.561	1.428×
	full32 continued@25k	25k [‡]	32	32	7.670	1.037×
	ShortGPT-16	200k	16	16	9.780	1.322×
1B	frozen-front control	200k	14	16	12.797	1.730×
	fully random-init control	200k	0	16	11.498	1.554×
	full base	—	16	16	10.642	1.000×
	keep7+fresh2	50k	7	9	17.619	1.656×
	keep7+fresh2	100k	7	9	16.161	1.519×
	keep7+fresh2	147k	7	9	15.628	1.469×

Table 4: **All held-out perplexity checkpoints used in this study.** Every point uses the same disjoint 4096×2048 Dolmino validation shard (8,384,512 target tokens) and token-weighted eight-shard merging. The frozen-front arm retains the same pretrained prefix but does not update it; the fully random-init arm has the same 16-layer shape but loads no pretrained weights. ShortGPT inherits non-contiguous layers [0–12, 16, 17, 31] and appends no fresh blocks. [†]The depth-ledger endpoint values come from the retained keep8 at 121k, keep10 at 83.5k, and keep12 at 124k checkpoints after plateau stopping; no literal step-200k evaluation is available for these arms, and PPL was still decreasing. [‡]The intact 32-layer control stopped at its pre-registered likelihood plateau; all reported full32 results use the step-25k checkpoint.

most other metrics move by less than one point. Table 14 additionally exposes raw and continuation-length-normalized scoring for BoolQ, CommonsenseQA, and SocialIQA, whose options have unequal lengths; the main tables consistently use raw accuracy for these tasks. Tables 16 and 17 report the complete MMLU interface control and the zero-shot closed-book results used in the main analysis. Table 15 uses the retained per-example predictions to compare final keep14 with fully random init on the same examples. All five differences favor keep14 under both exact McNemar and paired-bootstrap criteria; frozen-front is not included because its per-example predictions were not retained.

The confidence intervals in Table 12 distinguish the available frontier statistically: keep8, early keep10, and fully random init include chance, while keep12 and keep14 are above it. The subject-group table below shows that recovery is heterogeneous rather than concentrated in a single aggregate score.

ShortGPT point	PPL	core6	MMLU
step 0	401.124	—	.2620
step 200k	9.780	.6215	.4739

Table 5: **Non-contiguous ShortGPT endpoint.** Cosine block influence selects 16 of 32 pretrained layers, [0–12, 16, 17, 31], with no fresh tail. The unhealed model is initially degenerate; after the same 200k-step Dolmino healing budget it recovers 63.0% of base above-chance MMLU.

Task	metric	10k	25k	44k	121k	$\Delta_{44 \rightarrow 121}$
MMLU	raw	.2542	.2502	.2463	.2535	+0.72
HellaSwag	norm.	.3915	.4390	.4694	.5169	+4.75
ARC-Challenge	norm.	.2654	.3114	.3140	.3601	+4.61
ARC-Easy	norm.	.5581	.5930	.6103	.6519	+4.16
PIQA	norm.	.6632	.6801	.6937	.7165	+2.28
WinoGrande	raw	.5209	.5083	.5185	.5312	+1.27
OpenBookQA	norm.	.3000	.3360	.3320	.3660	+3.40
LAMBADA	raw	.3429	.3827	.4333	.4463	+1.31
BoolQ	raw	.5713	.6080	.5881	.5948	+0.67
CommonsenseQA	raw	.3923	.4103	.4423	.4521	+0.98
SocialIQA	raw	.3767	.3874	.4002	.3976	−0.26

Table 6: **Complete 11-task keep8 downstream trajectory.** The final column is the accuracy-point change from 44k to the retained 121k endpoint checkpoint under the pre-specified metric shown in column two; “norm.” denotes accuracy based on token-normalized candidate log-likelihood. MMLU remains near chance while most other tasks improve. Earlier PPL checkpoints use a separate grid; the final held-out PPL is reported at 121k in Table 4.

A.4 MMLU subject-group results

Table 18 quantifies the broad-domain pattern. The inherited keep14 model is strongest in social science and the heterogeneous “Other” group, whereas science, technology, engineering, and mathematics (STEM) and humanities remain closer to chance. Table 22, placed at the end of the appendix as two side-by-side panels, reports every one of the 57 subjects at the 153.5k and final 200k keep14 checkpoints.

B Evaluation and Reproducibility Details

B.1 Data, optimization, and architecture reconstruction

All training uses English-language OLMo-2/Dolmino artifacts. The OLMo-2 checkpoint and Dolmino release are used under their published research licenses; the evaluation datasets retain their upstream terms and are not redistributed with our artifacts. Experiments ran on 8-GPU H20 or B200 nodes depending on the arm. The principal keep14 run used 200k optimizer steps;

Model	step	HS	ARC-C	ARC-E	PIQA	WinoG	OBQA	PPL
1B full base	—	.683	.422	.735	.757	.643	.400	10.642
keep7+fresh2	50k	.411	.289	.568	.671	.531	.318	17.619
keep7+fresh2	100k	.438	.300	.593	.687	.526	.326	16.161
keep7+fresh2	147k	.450	.311	.595	.692	.530	.326	15.628
keep7+fresh2	148.5k	.448	.294	.606	.695	.544	.330	—

Table 7: **OLMo-2-1B core-task trajectory.** We report `acc_norm` (continuation-length-normalized likelihood accuracy) except for WinoGrande, where raw and normalized accuracy coincide. Abbreviations: HS = HellaSwag; ARC-C/E = ARC-Challenge/Easy; WinoG = WinoGrande; OBQA = OpenBookQA; PPL = perplexity.

Model	step	MMLU	LAMBADA	BoolQ	CSQA	SIQA
1B full base	—	.3820	.6480	.624	.553	.422
keep7+fresh2	50k	.2495	.3536	.618	.368	.384
keep7+fresh2	100k	.2558	.3740	.597	.380	.394
keep7+fresh2	147k	.2529	.4021	.584	.391	.384
keep7+fresh2	150k	.2480	.3969	.577	.387	.386

Table 8: **OLMo-2-1B knowledge/comprehension trajectory.** MMLU remains at the .25 chance floor across the complete measured trajectory, whereas LAMBADA and commonsense tasks improve. Raw accuracy is shown for these five tasks.

the full32 control stopped at its pre-registered 25k likelihood plateau, and shallower prefix arms use the literal retained checkpoints stated in the tables. Because exploratory and failed runs were not logged uniformly, we cannot provide an exact project-wide GPU-hour total. We therefore report per-run hardware and steps rather than an unreliable aggregate.

Healing uses the DataComp for Language Models (DCLM) portion of `dolmino-mix-1124`, tokenized into 2048-token windows with the OLMo-2 tokenizer. The training array is disjoint from the fixed validation shard of 4096 windows. All main arms use effective batch size 128 and a cosine schedule after 150 warmup steps. In the executed distributed runs, all parameters in inherited arms use peak/minimum learning rates $2 \times 10^{-5} / 2 \times 10^{-6}$; the fully random-init arm uses $10^{-4} / 10^{-5}$ for all parameters. We use AdamW with $\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$, weight decay 0.1 on matrix parameters and zero on vectors, and global gradient-norm clipping at 1.0. Parameters and optimizer states use 32-bit floating point (fp32), while forward passes use bfloat16 (bf16) automatic mixed precision; gradient checkpointing is enabled. The maximum is 200k optimizer steps. No BA-BILong or other synthetic task data is mixed into healing.

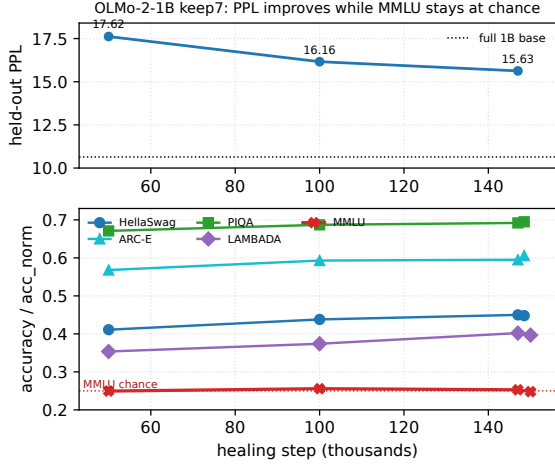


Figure 6: **Qualitative OLMo-2-1B trajectory.** PPL and several ordinary tasks improve while MMLU stays near chance over the observed keep7 interval.

Qwen3-8B arm	Depth	PPL↓	MMLU letter
Full base	36/36	11.42	.7297
keep12 + fresh2, 200k	14/36	23.49	.2495

Table 9: **Cross-family endpoint diagnostic.** The Qwen3-8B (Yang et al., 2025) arm heals on SlimPajama rather than Dolmino and is more compressed than the principal OLMo arm. It supports only the directional answer-letter finding; absolute PPL and recovery magnitude are not compared across families. This endpoint does not include the content-MMLU and closed-book controls used in the principal study.

Runs do not set an explicit random seed, so the exact fresh initialization and initial shuffle are not reproducible from the command line alone. The keep14 run resumed once from step 34.5k with model, optimizer, and pseudorandom-number generator states restored. The checkpoint did not store the within-epoch data-loader offset, so the resumed iterator restarted that epoch’s distributed shuffle; token counts in this paper are therefore nominal presentations rather than claims about unique examples.

For each inherited arm, launch-time checks require exact equality between the base and transplanted front blocks, embedding, final norm, and language-model (LM) vocabulary output head ($\max |\theta_{\text{arm}} - \theta_{\text{base}}| = 0$). Fresh-block norms are initialized to one and fresh query projections have standard deviation 0.02. Evaluation reconstructs the pruned architecture from checkpoint metadata and the saved parameter-state dictionary, then loads every tensor strictly. The keep14 models have

4.060B parameters.

B.2 Likelihood scoring and dataset sizes

For each multiple-choice candidate, the evaluator concatenates context and continuation, supplies the candidate tokens to the model, and sums their conditional log probabilities. This teacher-forced scoring never uses free generation. For the broad likelihood suite, `acc_norm` follows the task harness’s continuation-length convention. The separate MMLU content protocol reports both summed log-likelihood and mean log-likelihood per continuation token; the latter is the content headline. WinoGrande uses its standard cloze format, in which each candidate fills the blank in the supplied sentence. MMLU candidates are answer letters under a fixed subject-description prompt. LAMBADA reports whether the entire final-word continuation is the greedy token sequence.

Evaluation integrity. Perplexity is merged as $\exp(\sum_g \text{NLL}_g / \sum_g n_g)$ over eight shards, where NLL_g is shard g ’s summed negative log-likelihood and n_g is its token count; we never average per-shard perplexities. At keep14 steps 128k, 153.5k, and 200k, independent recomputation from all shard files reproduces every PPL and downstream cell to 10^{-9} . All evaluated task cells have zero NaNs and full sample counts. BoolQ contains the same two overlength passages in every run containing BoolQ; they are truncated consistently and still scored. No other task is truncated. The full 7B base reproduces expected OLMo-2 reference values (HellaSwag .805, MMLU .6053, LAMBADA .732), and the 1B base similarly reproduces its expected range.

C Mechanistic Probe Details

The MMLU logit lens uses 1,000 MMLU test questions. At each layerwise hidden-state depth, the final-prompt-position hidden state is passed through the model’s final normalization and output head; accuracy is the argmax among the letter tokens {A,B,C,D}. We define onset as the first layer exceeding chance by .05 and `sat95` as the first layer reaching 95% of the top-layer above-chance signal. On OLMo these occur at layers 18 and 19, while `sat99` is reached at layer 27. Complete OLMo and supplemental Qwen values appear in Table 21.

The new same-model OLMo probe adds frozen linear classifiers for WiC, SST-2, and RTE at every depth plus exact next-token top-1 agreement

Arm	step	MMLU	HS	ARC-C	ARC-E	PIQA	WinoG	OBQA	LAMB.	BoolQ	CSQA	SIQA
7B base full (32L)	—	.6053	.805	.571	.829	.811	.744	.462	.732	.815	.665	.502
keep8 (10L)	121k	.2535	.517	.360	.652	.717	.531	.366	.446	.595	.452	.398
keep10 (12L)	83.5k	.2718	.547	.366	.644	.726	.542	.356	.496	.609	.454	.415
keep12 (14L)	124k	.2752	.599	.394	.689	.737	.607	.376	.535	.587	.496	.410
keep14 (16L)	128k	.3012	.631	.426	.702	.747	.630	.402	.575	.639	.505	.423
keep14 (16L)	153.5k	.3124	.643	.442	.705	.745	.633	.406	.570	.606	.506	.441
keep14 (16L)	200k	.3191	.645	.438	.705	.745	.626	.404	.577	.638	.499	.434
ShortGPT-16	200k	.4739	.685	.476	.746	.758	.655	.408	.619	.729	.534	.442
frozen-front (16L)	200k	.2628	.595	.381	.666	.724	.646	.366	.513	.592	.453	.414
fully random-init (16L)	200k	.2461	.578	.414	.697	.733	.545	.384	.484	.614	.450	.416

Table 10: **Downstream zero-shot likelihood scoring (7B)**. `acc_norm` is used for HS, ARC-C/E, PIQA, and OBQA; raw accuracy is used for MMLU, WinoG, LAMBADA, BoolQ, CSQA, and SIQA. Here `acc_norm` means accuracy from continuation-length-normalized log-likelihood. Appendix Table 14 reports both raw and normalized views for the final three tasks. Abbreviations: HS = HellaSwag; ARC-C/E = ARC-Challenge/Easy; WinoG = WinoGrande; OBQA = OpenBookQA; LAMB. = LAMBADA; CSQA = CommonsenseQA; SIQA = SocialQA. The prefix ladder places keep8 at MMLU chance and final keep14 at 19.4% above-chance recovery. ShortGPT-16 reaches 63.0%, but also inherits two more pretrained blocks and the original final layer. Frozen-front recovers only 3.6%, while fully random init remains at chance despite nontrivial scores elsewhere and similar raw BoolQ accuracy to keep14 (.614 vs. .638). The shallow prefix ladder uses retained plateau checkpoints and is not step- or compute-matched; keep14, ShortGPT, and the same-shape controls are literal 200k evaluations.

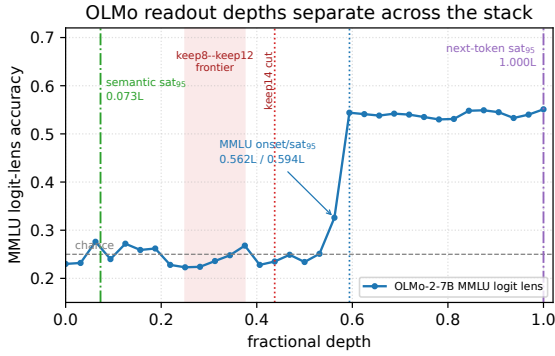


Figure 7: **Probe-dependent readout depths on OLMo-2-7B**. The shaded band marks prefix cuts used by the available behavioral points. Thresholds use different readouts and are descriptive; they are neither directly comparable causal boundaries nor locations of stored knowledge.

with the final layer. Their final-layer-relative sat_{95} depths are layers 1, 5, and 1, giving a mean of $2.33/32 = 0.073L$; because intermediate probes can outperform the final layer, this is a readout heuristic rather than a causal semantic boundary. Next-token agreement reaches sat_{95} only at layer 32 ($1.000L$). Together with MMLU sat_{95} at $0.594L$, this yields the same-model ordering semantic < MMLU < next-token. The keep14 step-200k evaluation is included throughout the main results and appendix.

Family	Task	chance	base	inherited	random init	recovery (%)
				reported score		inherited / random
surface	ARC-Easy	.25	.829	.705	.697	78.6/77.2
surface	PIQA	.50	.811	.745	.733	78.9/74.9
reasoning	HellaSwag	.25	.805	.645	.578	71.1/59.1
reasoning	ARC-Challenge	.25	.571	.438	.414	58.5/51.1
reasoning	WinoGrande	.50	.744	.626	.545	51.6/18.4
reasoning	OpenBookQA	.25	.462	.404	.384	72.6/63.2
reasoning	LAMBADA	.00	.732	.577	.484	78.9/66.1
reasoning	SocialQA	.333	.502	.434	.416	59.8/49.1
reasoning	CommonsenseQA	.20	.665	.499	.450	64.3/53.9
in-context	BoolQ	.50	.815	.638	.614	43.9/36.2
answer-letter	MMLU	.25	.6053	.3191	.2461	19.4/ - 1.1

Table 11: **Exact above-chance recovery for inherited keep14 and the fully random-init control at the matched step 200k.** Recovery is $100(x - c)/(b - c)$ for score x , chance c , and full-base score b . Family labels are descriptive: “surface” contains ARC-Easy and PIQA; “in-context” denotes passage-grounded BoolQ; “answer-letter” denotes the MMLU protocol; the remaining tasks are grouped descriptively as “reasoning.” Reported scores use continuation-length-normalized likelihood for HellaSwag, ARC, PIQA, and OpenBookQA and raw accuracy otherwise. The negative random-init MMLU value means statistically chance-level performance. The random-init arm also randomizes lexical modules and uses a different learning rate, so the table is an operating-point comparison rather than an initialization-only or decoder-block-only ablation. Negative recovery denotes a score below the chance-adjusted reference.

Arm	MMLU	SE	approx. 95% CI
full 7B base	.6053	.0041	[.5972, .6134]
keep8, 121k	.2535	.0037	[.2463, .2607]
keep10, 83.5k	.2718	.0038	[.2644, .2792]
keep12, 124k	.2752	.0038	[.2678, .2826]
keep14, 128k	.3012	.0039	[.2936, .3088]
keep14, 153.5k	.3124	.0039	[.3047, .3201]
keep14, 200k	.3191	.0039	[.3114, .3268]
ShortGPT-16, 200k	.4739	.0042	[.4656, .4822]
frozen-front, 200k	.2628	.0037	[.2555, .2701]
fully random init, 200k	.2461	.0036	[.2390, .2532]

Table 12: **Marginal MMLU uncertainty across the 7B frontier.** Standard errors and Wald intervals use $n = 14,042$ questions. keep8 and fully random init include the .25 chance floor; keep10/keep12, frozen-front, all keep14 checkpoints, and ShortGPT are above it. These marginal intervals summarize each arm separately; Table 15 gives the stronger item-paired three-control comparisons.

Metric	128k	153.5k	200k
held-out PPL	10.826	10.693	10.561
MMLU	.3012	.3124	.3191
HellaSwag	.631	.643	.645
ARC-Challenge	.426	.442	.438
ARC-Easy	.702	.705	.705
PIQA	.747	.745	.745
WinoGrande	.630	.633	.626
OpenBookQA	.402	.406	.404
LAMBADA	.575	.570	.577
BoolQ (raw acc.)	.639	.606	.638
CommonsenseQA (raw)	.505	.506	.499
SocialQA (raw)	.423	.441	.434

Table 13: **Complete late keep14 trajectory.** From 153.5k to 200k, held-out PPL improves by 0.132 while MMLU gains only 0.67 points. Over the paired 14,042-item rerun from 128k to 200k, MMLU increases by 1.68 points (95% CI [1.08, 2.29], exact McNemar $p = 4.12 \times 10^{-8}$). Most other metrics remain within one point from 153.5k to 200k, with mixed signs; continued healing yields a small but detectable answer-letter gain rather than broad late catch-up.

Task	metric	base	k8	k12	k14-p	k14-f	frozen	random
BoolQ	raw	.815	.588	.610	.606	.638	.592	.614
BoolQ	norm.	.753	.620	.657	.682	.689	.700	.619
CSQA	raw	.665	.442	.508	.506	.499	.453	.450
CSQA	norm.	.652	.405	.471	.479	.476	.467	.405
SIQA	raw	.502	.400	.415	.441	.434	.414	.416
SIQA	norm.	.547	.431	.460	.474	.474	.451	.443

Table 14: **Raw versus continuation-length-normalized accuracy.** The normalized score divides candidate log-likelihood by continuation character length before selecting an answer. The main results pre-specify raw accuracy for BoolQ, CommonsenseQA, and SocialIQA. Normalization can materially change BoolQ and SIQA, so both views are reported here. k14-p/k14-f denote keep14 at 153.5k/200k; frozen-front and random init are the other 200k controls.

Comparison	n	Δ (pp)	McNemar p	95% CI (pp)
keep14 – random init	14,042	+7.11	1.63×10^{-46}	[6.14, 8.09]
keep14 – frozen front	14,042	+5.50	6.99×10^{-27}	[4.51, 6.49]
frozen front – random init	14,042	+1.61	2.59×10^{-3}	[0.58, 2.64]

Table 15: **Paired MMLU comparisons among the three 200k, 16-layer operating points.** Δ follows the row order and is computed from the aligned per-item rerun (keep14 .31797, frozen .26293, random .24683), which differs slightly from the separately rounded aggregate cells. McNemar uses an exact two-sided binomial test on discordant correctness pairs; 95% confidence intervals use 10,000 paired bootstrap resamples with seed 1234. All intervals exclude zero.

Arm	letter	content raw	content norm.
Full base	.6054	.4400	.4706
Full32 at 25k	.5877	.4344	.4662
keep8 endpoint [†]	.2550	.3219	.3423
keep10 endpoint [†]	.2720	.3230	.3445
keep12 endpoint [†]	.2728	.3419	.3629
keep14 at 200k	.3184	.3548	.3832
Frozen front at 200k	.2624	.3349	.3604
Random init at 200k	.2470	.3439	.3598
ShortGPT-16 at 200k	.4742	.3764	.4012

Table 16: **MMLU scoring-interface control on all 14,042 items.** Letter scores one of A/B/C/D after displaying the options. Content scores the complete option text. “Content raw” selects by summed continuation log-likelihood; “content norm.” selects by mean log-likelihood per continuation token and is the pre-specified content headline. [†]The ledger endpoint labels refer to retained keep8 at 121k, keep10 at 83.5k, and keep12 at 124k checkpoints; no literal step-200k content-protocol evaluations are available for these arms.

Arm	PopQA contains	TriviaQA EM	NQ EM
Full base	.2571	.6355	.2050
Full32 at 25k	.2280	.5715	.1582
keep14 at 200k	.1415	.2940	.0598
Frozen front at 200k	.1283	.2477	.0496
Random init at 200k	.1112	.2086	.0632
keep8 endpoint [†]	.1246	.1577	.0368

Table 17: **Zero-shot, no-retrieval closed-book evaluation.** All arms use the same plain Question:/Answer: generation protocol. All full32 cells use the intact 32-layer step-25k checkpoint. [†]keep8 has a different 10-layer shape; its retained checkpoint is step 121k and it is included only as a depth reference.

Group	n	base	full32	keep14	ShortGPT	random
STEM	3,864	.534	.507	.294	.405	.236
Humanities	4,705	.555	.542	.299	.428	.252
Social science	3,077	.709	.690	.335	.554	.240
Other	2,396	.686	.670	.377	.574	.259

Table 18: **Sample-weighted MMLU broad-group accuracy.** full32 is the 25k continued-full-model control; keep14, ShortGPT, and random are their reported 200k endpoints. Group definitions follow the standard four-way MMLU taxonomy.

Task	split / mode	n	chance	metric
HellaSwag	validation, 4-choice	10,042	.25	acc_norm
ARC-Challenge	test, 4-choice	1,172	.25	acc_norm
ARC-Easy	test, 4-choice	2,376	.25	acc_norm
PIQA	validation, 2-choice	1,838	.50	acc_norm
WinoGrande	validation, 2-choice cloze	1,267	.50	accuracy
OpenBookQA	test, 4-choice	500	.25	acc_norm
MMLU	test, 57 subjects, 4-choice letters	14,042	.25	accuracy
LAMBADA-OpenAI	test, greedy final word	5,153	≈ 0	exact greedy accuracy
BoolQ	validation, yes/no	3,270	.50	accuracy
CommonsenseQA	validation, 5-choice	1,221	.20	accuracy
SocialIQA	validation, 3-choice	1,954	.333	accuracy

Table 19: **Evaluation suite and chance floors.** All tasks are zero-shot and likelihood scored; no free-form generation or chat template is used. acc_norm selects answers after dividing candidate log-likelihood by continuation character length. All listed examples are evaluated.

OLMo-2-7B quantity	layer	fractional depth
semantic probe mean sat ₉₅	2.33	0.073 L
MMLU onset (chance+0.05)	18	0.562 L
MMLU sat ₉₅	19	0.594 L
next-token sat ₉₅	32	1.000 L

Table 20: **Three ordered readout depths on the same OLMo model.** Semantic sat₉₅ averages WiC, SST-2, and RTE depths measured relative to each final-layer probe score. MMLU uses the final norm and output head; next-token sat₉₅ is exact top-1 agreement with the final layer. These are readout thresholds, not causal boundaries.

(a) OLMo-2-7B								(b) Qwen3-8B							
L	frac.	acc.	gold LL	L	frac.	acc.	gold LL	L	frac.	acc.	gold LL	L	frac.	acc.	gold LL
0	0.0000	0.230	-14.499	17	0.5312	0.251	-11.947	0	0.0000	0.272	-22.017	19	0.5278	0.237	-13.084
1	0.0312	0.232	-14.024	18	0.5625	0.326	-11.652	1	0.0278	0.234	-24.014	20	0.5556	0.237	-12.790
2	0.0625	0.276	-13.244	19	0.5938	0.544	-8.517	2	0.0556	0.240	-24.741	21	0.5833	0.236	-12.816
3	0.0938	0.240	-12.939	20	0.6250	0.541	-8.038	3	0.0833	0.236	-24.112	22	0.6111	0.235	-10.357
4	0.1250	0.272	-13.112	21	0.6562	0.538	-7.002	4	0.1111	0.233	-22.029	23	0.6389	0.232	-10.675
5	0.1562	0.259	-12.793	22	0.6875	0.542	-5.694	5	0.1389	0.237	-17.364	24	0.6667	0.236	-9.027
6	0.1875	0.262	-12.206	23	0.7188	0.540	-6.038	6	0.1667	0.242	-12.939	25	0.6944	0.621	-6.590
7	0.2188	0.228	-11.715	24	0.7500	0.535	-4.847	7	0.1944	0.234	-12.999	26	0.7222	0.625	-6.928
8	0.2500	0.223	-11.116	25	0.7812	0.530	-5.091	8	0.2222	0.229	-13.867	27	0.7500	0.624	-6.054
9	0.2812	0.224	-11.294	26	0.8125	0.531	-5.701	9	0.2500	0.241	-14.885	28	0.7778	0.635	-6.225
10	0.3125	0.236	-11.493	27	0.8438	0.548	-5.345	10	0.2778	0.237	-14.034	29	0.8056	0.572	-6.685
11	0.3438	0.248	-12.051	28	0.8750	0.549	-5.456	11	0.3056	0.229	-14.021	30	0.8333	0.629	-7.052
12	0.3750	0.268	-11.847	29	0.9062	0.545	-5.402	12	0.3333	0.228	-13.218	31	0.8611	0.630	-7.164
13	0.4062	0.228	-11.499	30	0.9375	0.533	-5.338	13	0.3611	0.225	-13.516	32	0.8889	0.629	-7.869
14	0.4375	0.235	-11.295	31	0.9688	0.540	-4.475	14	0.3889	0.229	-13.597	33	0.9167	0.629	-7.898
15	0.4688	0.249	-11.555	32	1.0000	0.551	-3.791	15	0.4167	0.231	-13.184	34	0.9444	0.638	-7.334
16	0.5000	0.234	-11.960					16	0.4444	0.237	-14.466	35	0.9722	0.636	-4.900
								17	0.4722	0.236	-14.439	36	1.0000	0.632	-5.412
								18	0.5000	0.231	-13.220				

Table 21: **Complete MMLU logit-lens trajectories.** Layer 0 is the embedding/input residual state and layer L follows decoder block L . Accuracy is four-letter MMLU accuracy; gold LL is full-vocabulary gold-letter log-likelihood on the same 1,000 questions. These are output-head readouts, not causal storage localizations.

Subject	n	base	k8	k12	k14-p	k14-f	frozen	rand	Subject	n	base	k8	k12	k14-p	k14-f	frozen	rand
abstract_algebra	100	.310	.300	.240	.260	.270	.310	.240	high_school_statistics	216	.514	.171	.338	.204	.213	.218	.194
anatomy	135	.622	.289	.296	.363	.363	.296	.215	high_school_us_history	204	.824	.314	.230	.319	.304	.299	.309
astronomy	152	.730	.171	.289	.316	.322	.283	.178	high_school_world_history	237	.797	.253	.270	.325	.354	.274	.253
business_ethics	100	.590	.250	.340	.390	.430	.260	.220	human_aging	223	.650	.283	.139	.368	.359	.265	.341
clinical_knowledge	265	.649	.242	.275	.317	.317	.257	.249	human_sexuality	131	.756	.221	.298	.382	.382	.298	.321
college_biology	144	.771	.278	.271	.347	.326	.250	.264	international_law	121	.777	.248	.240	.281	.264	.355	.240
college_chemistry	100	.470	.190	.310	.270	.230	.260	.200	jurisprudence	108	.685	.204	.269	.287	.315	.231	.259
college_computer_science	100	.490	.240	.300	.300	.290	.240	.270	logical_fallacies	163	.693	.233	.215	.380	.350	.209	.221
college_mathematics	100	.360	.200	.310	.240	.320	.270	.230	machine_learning	112	.455	.170	.277	.375	.330	.277	.321
college_medicine	173	.566	.277	.341	.295	.289	.249	.214	management	103	.777	.204	.359	.320	.369	.282	.194
college_physics	102	.422	.196	.304	.225	.275	.265	.147	marketing	234	.825	.286	.274	.440	.440	.286	.312
computer_security	100	.680	.230	.250	.410	.400	.340	.290	medical_genetics	100	.740	.170	.300	.370	.400	.340	.320
conceptual_physics	235	.570	.238	.264	.311	.332	.243	.277	miscellaneous	783	.803	.271	.278	.436	.457	.268	.244
econometrics	114	.360	.246	.193	.246	.246	.325	.237	moral_disputes	346	.694	.225	.260	.329	.321	.272	.237
electrical_engineering	145	.600	.255	.276	.359	.366	.345	.248	moral_scenarios	895	.279	.238	.238	.244	.253	.244	.238
elementary_mathematics	378	.397	.278	.254	.296	.272	.259	.222	nutrition	306	.693	.239	.307	.324	.327	.320	.242
formal_logic	126	.389	.302	.325	.198	.262	.183	.302	philosophy	311	.711	.235	.270	.322	.318	.257	.222
global_facts	100	.360	.340	.330	.280	.290	.320	.190	prehistory	324	.741	.318	.275	.346	.358	.299	.238
high_school_biology	310	.765	.223	.319	.319	.319	.239	.190	professional_accounting	282	.415	.266	.245	.238	.241	.262	.280
high_school_chemistry	203	.493	.261	.246	.256	.232	.202	.167	professional_law	1534	.460	.244	.246	.259	.272	.245	.263
high_school_computer_science	100	.590	.290	.220	.300	.290	.280	.290	professional_medicine	272	.570	.199	.239	.228	.232	.301	.301
high_school_european_history	165	.752	.236	.236	.242	.309	.194	.206	professional_psychology	612	.611	.243	.247	.320	.302	.260	.275
high_school_geography	198	.783	.242	.308	.384	.328	.288	.207	public_relations	110	.627	.282	.236	.291	.327	.309	.236
high_school_government_and_politics	193	.870	.202	.285	.368	.358	.244	.197	security_studies	245	.727	.367	.339	.294	.335	.245	.245
high_school_macroeconomics	390	.636	.226	.338	.295	.321	.285	.221	sociology	201	.836	.289	.279	.418	.428	.194	.294
high_school_mathematics	270	.296	.230	.278	.281	.248	.259	.219	us_foreign_policy	100	.850	.230	.340	.480	.450	.270	.340
high_school_microeconomics	238	.655	.239	.269	.277	.277	.252	.185	virology	166	.524	.241	.313	.355	.386	.283	.313
high_school_physics	151	.404	.238	.358	.225	.245	.298	.212	world_religions	171	.836	.246	.292	.491	.509	.287	.322
high_school_psychology	545	.807	.196	.297	.317	.358	.229	.207									

Table 22: **Complete 57-subject MMLU results.** All values are accuracy. k14-p and k14-f are inherited keep14 at 153.5k and 200k; frozen-front and random init are the two other 200k controls. Subjects are split alphabetically across the two panels.